

Dirichlet Integral

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0 Introduction

Lemma 0.1 (Dirichlet Kernel). For all $n \in \mathbb{N}_0$ and all $x \in \mathbb{R}$, we have:

$$1 + 2 \sum_{k=1}^n \cos(2kx) = \frac{\sin((2n+1)x)}{\sin(x)}$$

Proof. Clearly the identity holds for $n = 0$ as both sides are 1. Suppose the identity holds for some $n \in \mathbb{N}_0$. We want to show it holds for $n + 1$. We have:

$$\begin{aligned} 1 + 2 \sum_{k=1}^{n+1} \cos(2kx) &= 1 + 2 \sum_{k=1}^n \cos(2kx) + 2 \cos((2n+2)x) = \frac{\sin((2n+1)x)}{\sin(x)} + 2 \cos((2n+2)x) \\ &= \frac{\sin((2n+1)x) + 2 \sin(x) \cos((2n+2)x)}{\sin(x)} \end{aligned}$$

Note that $\sin((2n+3)x) - \sin((2n+1)x) = 2 \sin(x) \cos((2n+2)x)$. This yields:

$$1 + 2 \sum_{k=1}^{n+1} \cos(2kx) = \frac{\sin((2n+1)x) + \sin((2n+3)x) - \sin((2n+1)x)}{\sin(x)} = \frac{\sin((2n+3)x)}{\sin(x)}$$

Thus the identity holds for $n + 1$, and so it holds for all $n \in \mathbb{N}_0$. 🤖

Theorem 0.2 (Dirichlet Integral).

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

Proof. Define $K : [0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ as follows:

$$K(x) = \begin{cases} \frac{\sin((2n+1)x)}{\sin(x)} & x \neq 0 \\ 2n+1 & x = 0 \end{cases}$$

Clearly K is continuous on $(0, \frac{\pi}{2}]$, and:

$$\begin{aligned} \lim_{x \rightarrow 0^+} K(x) &= \lim_{x \rightarrow 0^+} \frac{\sin((2n+1)x)}{\sin(x)} \\ &= \lim_{x \rightarrow 0^+} (2n+1) \frac{\sin((2n+1)x)}{(2n+1)x} \frac{x}{\sin(x)} \\ &= (2n+1) \left(\lim_{x \rightarrow 0^+} \frac{\sin((2n+1)x)}{(2n+1)x} \right) \left(\lim_{x \rightarrow 0^+} \frac{x}{\sin(x)} \right) \\ &= (2n+1) \cdot 1 \cdot 1 \\ &= 2n+1 \end{aligned}$$

Thus K is continuous (and thus Riemann integrable) on $[0, \frac{\pi}{2}]$. Using the Dirichlet kernel, we have:

$$\int_0^{\pi/2} \frac{\sin((2n+1)x)}{\sin(x)} dx = \int_0^{\pi/2} \left(1 + 2 \sum_{k=1}^n \cos(2kx) \right) dx = \int_0^{\pi/2} 1 dx + 2 \sum_{k=1}^n \int_0^{\pi/2} \cos(2kx) dx = \frac{\pi}{2}$$

We also have:

$$\int_0^{\pi/2} \frac{\sin((2n+1)x)}{x} dx = \int_0^{(2n+1)\pi/2} \frac{\sin(x)}{x} dx$$

As $n \rightarrow \infty$, this tends to $\int_0^\infty \frac{\sin(x)}{x} dx$. Define $g : (0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ by $g(x) = \frac{1}{x} - \frac{1}{\sin(x)}$. Clearly g is continuous on $(0, \frac{\pi}{2}]$.

Also, using L'Hôpital's rule twice, we have:

$$\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} \frac{\sin(x) - x}{x \sin(x)} = \lim_{x \rightarrow 0^+} \frac{\cos(x) - 1}{\sin(x) + x \cos(x)} = \lim_{x \rightarrow 0^+} \frac{-\sin(x)}{2 \cos(x) - x \sin(x)} = \frac{0}{2 - 0} = 0$$

Thus g can be made continuous on $[0, \frac{\pi}{2}]$ by setting $g(0) = 0$. By the Riemann-Lebesgue lemma, we have $\int_0^{\pi/2} \left(\frac{1}{x} - \frac{1}{\sin(x)} \right) \sin((2n+1)x) dx \rightarrow 0$. 😎

Lemma 0.3. Suppose $f \in C[0, \pi]$. Define:

$$b_n = \frac{2}{\pi} \int_0^\pi f(x) \sin(nx) dx \qquad S_n = \int_0^\pi \left(f(x) - \sum_{k=1}^n b_k \sin(kx) \right)^2 dx$$

Then:

$$S_n = \int_0^\pi f(x)^2 dx - \frac{2}{\pi} \sum_{k=1}^n b_k^2$$